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12.4. TWO-PHASE FERRITE-POLYMER COMPOSITES AND THREE-PHASE FERRITE-Fe-POLYMER COMPOSITES

Table 12.4 shows the parametric fits of susceptibility and permittivity for various two-phased ferrite polymer composites and three-phase ferrite-Fe-polymer composites. All samples were made in the laboratory and formed into coaxial sample shapes for testing. The functional forms are shown at the beginning of the table. *Note, use the absolute parameter values (i.e., |B|, |C|, |D|) when calculating permeability.* Functions are those causal functions that have been derived in **Chaps. 2** and **3** for dielectric and magnetic media. All data are based on laboratory-prepared samples and their measurement. *Caution should be observed when the derived parameters are over a narrow bandwidth that does not fully encompass the fundamental resonance. In those cases, extrapolation of the parameters outside the original measured band can lead to an error.* Chemical and/or physical composition is shown when available. In the table exponentials are often shown as bE—which means b*10^{-a}. Thus –2E-004–7.6E-003j is equal to –0.0002 –j0.0076. **Figure 12.6** shows a typical three-phase measurement and numerical fit and **Fig. 12.7** illustrates a fit to a two-phase composite.

Table 12.4 *Micrometer and Nanometer Ferrite and Ferrite-Fe Particulates and Composites*

Sample ID and Information $\mu(f)-1=\chi_m(f)=B(1-j*f/D)\{1-(f/C)^2-j*(f/D)\}^{-1}$ $\chi_{DCP}, f_{rc}, f_{rp}, f_{cd}, f_{pd}$ are, respectively, the EMT-calculated DC magnetic particle susceptibility; calculated composite resonant frequency, |*C*|; EMT-calculated magnetic particle resonant frequency; calculated composite relaxation frequency, |*D*|; EMT-calculated magnetic particle relaxation frequency. Note, please apply the absolute parameter values (i.e., |*B*|, |*C*|, |*D*|) when calculating permeability. $\varepsilon_1(f)=B+Cf^D+E\{1-(f/F)^2-2j*(f/G)\}^{-1}$ $\varepsilon_2(f)=B+real(C)f^D+imag(C)f^E+F\{1-(f/G)^2-2j*(f/H)\}^{-1}$	B	C	D	E	F	G	H
Composites of polymer and $Ni_{.32}Zn_{.48}Cu_{.12}Fe_{.2}O_{4.6}$							
About 80 µm 9%/volume in epoxy, 1.38 g/cc, 0.05–18 GHz,							
$\chi_m(f)$	0.33+ 0.1j	1.61+ 0.95j	–0.74– 0.4j				
|*B*|, |*C*| and |*D*|	0.34	1.8694	0.8412				
$\chi_{DCP}, f_{rc}, f_{cd}, f_{pd}$	62.3	1.8694	0.42	0.8412	0.04		

Sample ID and Information

$$\mu(f)-1=\chi_m(f)=B(1-j*f/D)\{1-(f/C)^2-j*(f/D)\}^{-1}$$

$\chi_{DCP}, f_{rc}, f_{rp}, f_{cd}, f_{pd}$ are, respectively, the EMT-calculated DC magnetic particle susceptibility; calculated composite resonant frequency, |*C*|; EMT-calculated magnetic particle resonant frequency; calculated composite relaxation frequency, |*D*|; EMT-calculated magnetic particle relaxation frequency. Note, please apply the absolute parameter values (i.e., |*B*|, |*C*|, |*D*|) when calculating permeability.

$$\varepsilon_1(f)=B+Cf^D+E\{1-(f/F)^2-2j*(f/G)\}^{-1}$$

$$\varepsilon_2(f)=B+real(C)f^D+imag(C)f^E+F\{1-(f/G)^2-2j*(f/H)\}^{-1}$$

	B	C	D	E	F	G	H
$\varepsilon_2(f)$	2.47+ 0.26j	8E-002– 0.14j	0.61+ 0.68j	–0.17+ 0.48j	0.66– 7E-002j	10.66– 12.9j	–18.6+ 10.47j
About 80 µ, m16%/volume in epoxy, 1.6 g/cc, 0.001–10 GHz							
$\chi_m(f)$	1.16+ 3E-002j	0.65+ 9E-002j	–0.2+ 3E-002j				
|*B*|, |*C*| and |*D*|	1.16	0.6562	0.2022				
$\chi_{DCP}, f_{rc}, f_{cd}, f_{pd}$		0.6562		0.2022			
$\varepsilon_2(f)$	2.66+ 0.16j	7E-002– 9E-002j	0.61+ 0.39j	0.26+ 0.42j	0.81+ 0.12j	4.3+ 10.9j	51.6+ 43.9j
About 80 µm 16.7%/volume in epoxy, 1.68 g/cc, 0.05–10 GHz							
$\chi_m(f)$	0.85+ 0.89j	0.38+ 1.45j	0.37– 0.45j	–0.76+ 1.03j			
|*B*|, |*C*| and |*D*|	1.23	1.499	0.583				
$\chi_{DCP}, f_{rc}, f_{cd}, f_{pd}$		1.499		0.583			
$\varepsilon_1(f)$	1.57+ 5E-002j	9E-002+ 0.14j	–1.7– 0.25j	1.71+ 0.12j	5E-002– 14.7j	1.52+ 43.6j	
About 80 µm 29.9%/volume in epoxy, 2.22 g/cc, 0.05–10 GHz,							
$\chi_m(f)$	2.01+ 0.91j	0.28– 0.82j	0.51– 0.79j	–6E-002– 0.79j			
|*B*|, |*C*| and |*D*|	2.21	0.866	0.940				
$\chi_{DCP}, f_{rc}, f_{cd}, f_{pd}$	31.8	0.87	0.19	0.55	0.07		

Sample ID and Information

$$\mu(f)-1=\chi_m(f)=B(1-j*f/D)\{1-(f/C)^2-j*(f/D)\}^{-1}$$

$\chi_{DCP}, f_{rc}, f_{rp}, f_{cd}, f_{pd}$ are, respectively, the EMT-calculated DC magnetic particle susceptibility; calculated composite resonant frequency, |*C*|; EMT-calculated magnetic particle resonant frequency; calculated composite relaxation frequency, |*D*|; EMT-calculated magnetic particle relaxation frequency. Note, please apply the absolute parameter values (i.e., |*B*|, |*C*|, |*D*|) when calculating permeability.

$$\varepsilon_1(f)=B+Cf^D+E\{1-(f/F)^2-2j*(f/G)\}^{-1}$$
$$\varepsilon_2(f)=B+real(C)f^D+imag(C)f^E+F\{1-(f/G)^2-2j*(f/H)\}^{-1}$$

	B	C	D	E	F	G	H
$\varepsilon_1(f)$	2.57+0.55j	1.46+0.56j	–3E-003–4E-003j	–0.44–1.1j	20.16+7.5j	–11.9–73.9j	
About 80 µm 29.9%/volume in epoxy, 2.22 g/cc, 0.001–18 GHz							
$\chi_m(f)$	2.04+7E-002j	0.53–0.16j	–0.17+0.17j	–0.98–0.53j			
|*B*|, |*C*| and |*D*|	2.04	0.554	0.240				
$\chi_{DCP}, f_{rc}, f_{cd}, f_{pd}$	27.2	0.56	0.21	0.24	0.03		
$\varepsilon_1(f)$	1.18+4E-002j	1.14+3E-002j	–9E-002+5E-002j	1.26+4E-002j	13.8–13.7j	6.03+90.4j	
About 80 µm 41.1%/volume in epoxy, 2.68 g/cc, 0.05–10 GHz							
$\chi_m(f)$	3.95+1.27j	0.37–0.48j	8E-002+0.27j	–0.61–0.89j			
|*B*|, |*C*| and |*D*|	4.15	0.61	0.28				
$\chi_{DCP}, f_{rc}, f_{cd}, f_{pd}$	29.3	0.61	0.11	0.28	0.042		
$\varepsilon_1(f)$	1.75+7E-002j	1.58+0.18j	–0.23+3E-002j	1.85+0.16j	14.68–10.64j	–34.2+15.8j	
$\varepsilon_2(f)$	3.23+0.44j	7E-002+0.11j	0.65+0.42j	–0.95+0.1j	1.96+0.17j	7.53–3.33j	–10.26–4.8j
About 80 µm 49.4%/volume in epoxy, about 3.0 g/cc, 0.001–10 GHz							
$\chi_m(f)$	6.18+0.15j	0.33-2E–002j	–0.11+5E-002j	–1.3–0.51j			
|*B*|, |*C*| and |*D*|	6.18	0.33	0.12				

Sample ID and Information							
$\mu(f)-1=\chi_m(f)=B(1-j*f/D)\{1-(f/C)^2-j*(f/D)\}^{-1}$							
χ_{DCP}, f_{rc}, f_{rp}, f_{cd}, f_{pd} are, respectively, the EMT-calculated DC magnetic particle susceptibility; calculated composite resonant frequency, |C|; EMT-calculated magnetic particle resonant frequency; calculated composite relaxation frequency, |D|; EMT-calculated magnetic particle relaxation frequency. Note, please apply the absolute parameter values (i.e., |B|, |C|, |D|) when calculating permeability.							
$\varepsilon_1(f)=B+Cf^D+E\{1-(f/F)^2-2j*(f/G)\}^{-1}$ $\varepsilon_2(f)=B+real(C)f^D+imag(C)f^E+F\{1-(f/G)^2-2j*(f/H)\}^{-1}$							
	B	C	D	E	F	G	H
χ_{DCP}, f_{rc}, f_{cd}, f_{pd}	29.8	0.33	0.133	0.12	0.02		
$\varepsilon_2(f)$	3.16+0.26j	0.2+8E-002j	0.75+0.52j	–0.2+0.2j	1.86+2E-002j	9.42–6.54j	–13.73–1.87j
About 80 µm 53.6% in epoxy, about 3.2 g/cc, 0.05–10 GHz							
$\chi_m(f)$	3.91+2.64j	0.23–0.68j	0.33+9E-002j	5E-002–1.06j			
|B|, |C| and |D|	4.72	0.7178	0.3421				
χ_{DCP}, f_{rc}, f_{cd}, f_{pd}	18.1	0.72	0.17	0.34	0.09		
$\varepsilon_1(f)$	1.63–2E-002j	1.47+7E-002j	–5E-002–0.17j	1.89-2E-002j	1.64+13.67j	14.2+22.1j	
$\varepsilon_2(f)$	3.28+2E-002j	0.31–9E-002j	0.39+0.3j	–0.94+0.63j	1.72–2E-003j	7.89+9.44j	6.18–49.19j
About 80 µm 58.4%/volume in epoxy, about 3.29 g/cc, 0.05–10 GHz							
$\chi_m(f)$	8.1+3.6j	0.12–0.32j	9E-002+8E-002j	–0.54–1.31j			
|B|, |C| and |D|	8.86	0.34	0.12				
χ_{DCP}, f_{rc}, f_{cd}, f_{pd}	31	0.34	0.05	0.12	0.02		
$\varepsilon_1(f)$	1.2–1.4j	1.9–0.98j	–0.12–0.16j	2.05+2.73j	7.32–15.6j	–33.7+20.6j	
$\varepsilon_2(f)$	3.49+0.18j	0.6–0.16j	0.34–3.7E-002j	–0.74+0.72j	2.29+2E-002j	2.87+2.01j	0.9–2.7j
About 80 µm 61.4%/volume in epoxy, about 3.52 g/cc, 0.05–10 GHz							

Sample ID and Information							
$\mu(f)-1=\chi_m(f)=B(1-j*f/D)\{1-(f/C)^2-j*(f/D)\}^{-1}$ $\chi_{DCP}, f_{rc}, f_{rp}, f_{cd}, f_{pd}$ are, respectively, the EMT-calculated DC magnetic particle susceptibility; calculated composite resonant frequency, |*C*|; EMT-calculated magnetic particle resonant frequency; calculated composite relaxation frequency, |*D*|; EMT-calculated magnetic particle relaxation frequency. Note, please apply the absolute parameter values (i.e., |*B*|, |*C*|, |*D*|) when calculating permeability. $\varepsilon_1(f)=B+Cf^D+E\{1-(f/F)^2-2j*(f/G)\}^{-1}$ $\varepsilon_2(f)=B+real(C)f^D+imag(C)f^E+F\{1-(f/G)^2-2j*(f/H)\}^{-1}$							
	B	C	D	E	F	G	H
$\chi_m(f)$	9.55+ 4.8j	0.16– 0.37j	0.13+ 0.12j	–0.38– 1.04j			
|*B*|, |*C*| and |*D*|	10.69	0.403	0.18				
$\chi_{DCP}, f_{rc}, f_{cd}, f_{pd}$	34.6	0.403	0.08	0.18	0.03		
$\varepsilon_1(f)$	1.89– 1.31j	2.63– 0.85j	–0.12– 0.18j	2.72+ 2.88j	14.49– 13.49j	–48.4+ 18.1j	
$\varepsilon_2(f)$	3.96+ 0.33j	1.05– 0.11j	9E-002– 0.2j	–1.02+ 0.82j	2.9+ 0.4j	8.7– 2.2j	–3.12– 11.9j
About 80 µm 61.8%/volume in epoxy, about 3.53 g/cc, 0.05–10 GHz							
$\chi_m(f)$	6.55+ 3.65j	0.24– 0.49j	0.18+ 0.19j	–0.37– 0.98j			
|*B*|, |*C*| and |*D*|	7.50	0.55	0.26				
$\chi_{DCP}, f_{rc}, f_{cd}, f_{pd}$	23.1	0.55	0.13	0.26	0.07		
$\varepsilon_1(f)$	1.88– 1.35j	2.1– 0.62j	–8E-002– 0.22j	2.54+ 2.62j	15.88– 23.4j	–54.1+ 17.5j	
$\varepsilon_2(f)$	3.85+ 0.25j	0.95- 0.15j	0.33– 0.28j	–1.04+ 0.65j	2.79+ 0.14j	7.3– 0.23j	1.48– 5.1j
About 80 µm 76.1% by volume in epoxy, about 4.12 g/cc, 0.001–10 GHz							
$\chi_m(f)$	14.2– 2E-002j	0.18– 1E-002j	–5E-002+ 1.6E-002j	–1.7– 0.37j			
|*B*|, |*C*| and |*D*|	14.20	0.18	0.053				
$\chi_{DCP}, f_{rc}, f_{cd}, f_{pd}$	31.7	0.18	0.03	0.053	0.014		

Sample ID and Information

$$\mu(f)-1=\chi_m(f)=B(1-j*f/D)\{1-(f/C)^2-j*(f/D)\}^{-1}$$

$\chi_{DCP}, f_{rc}, f_{rp}, f_{cd}, f_{pd}$ are, respectively, the EMT-calculated DC magnetic particle susceptibility; calculated composite resonant frequency, |*C*|; EMT-calculated magnetic particle resonant frequency; calculated composite relaxation frequency, |*D*|; EMT-calculated magnetic particle relaxation frequency. Note, please apply the absolute parameter values (i.e., |*B*|, |*C*|, |*D*|) when calculating permeability.

$$\varepsilon_1(f)=B+Cf^D+E\{1-(f/F)^2-2j*(f/G)\}^{-1}$$
$$\varepsilon_2(f)=B+real(C)f^D+imag(C)f^E+F\{1-(f/G)^2-2j*(f/H)\}^{-1}$$

	B	C	D	E	F	G	H
$\varepsilon_1(f)$	3.31+0.11j	2.32+0.15j	5E-003–6E-003j	1.87–0.22j	12.1–0.76j	–2.2–19.8j	
$\varepsilon_2(f)$	3.93+2E-002j	0.97–0.13j	3E-002+0.15j	–0.26+0.32j	2.8+0.16j	13.1–2.91j	–11.8–19j
Iron–Ferrite Composites 0.5							
500-nm Fe 7% + 80-µm $Ni_{.32}Zn_{.48}Cu_{.12}Fe_2O_{4.6}$ at 7%/volumes in epoxy, 1.8 g/cc, 0.001–18 GHz							
$\chi_m(f)$	1.04+0.15j	3.26–0.62j	–1.63+0.39j	–0.9+0.41j			
|*B*|, |*C*| and |*D*|	1.05	3.32	1.68				
$\chi_{DCP}, f_{rc}, f_{cd}, f_{pd}$		3.32		1.68			
$\varepsilon_1(f)$	0.99–1.45j	2.26–1.16j	–9.7E-003–2E-002j	2.1+2.73j	48.8+112j	47.2+119j	
$\varepsilon_2(f)$	2.94+0.29j	0.96–0.71j	0.49+8E-002j	0.32–0.12j	2.45+8E-002j	12.6+4j	–8.1–10.9j
500-nm Fe 10.2% + 80-µm $Ni_{.32}Zn_{.48}Cu_{.12}Fe_2O_{4.6}$ at 10.1%/volumes in epoxy, 2.1 g/cc, 0.001–10 GHz							
$\chi_m(f)$	0.98+0.13j	3.45+2.25j	–1.32–0.75j	–1.36–9E-002j			
|*B*|, |*C*| and |*D*|	0.99	4.12	1.52				
$\chi_{DCP}, f_{rc}, f_{cd}, f_{pd}$	27	4.12	0.19	1.52	0.19		
$\varepsilon_1(f)$ 1	0.97–1.54j	2.29+1.25j	0.11-6E–002j	2.1+2.69j	8.39+13.31j	18.7+15.7j	

Sample ID and Information

$$\mu(f)-1=\chi_m(f)=B(1-j*f/D)\{1-(f/C)^2-j*(f/D)\}^{-1}$$

$\chi_{DCP}, f_{rc}, f_{rp}, f_{cd}, f_{pd}$ are, respectively, the EMT-calculated DC magnetic particle susceptibility; calculated composite resonant frequency, $|C|$; EMT-calculated magnetic particle resonant frequency; calculated composite relaxation frequency, $|D|$; EMT-calculated magnetic particle relaxation frequency. Note, please apply the absolute parameter values (i.e., $|B|$, $|C|$, $|D|$) when calculating permeability.

$$\varepsilon_1(f)=B+Cf^D+E\{1-(f/F)^2-2j*(f/G)\}^{-1}$$

$$\varepsilon_2(f)=B+real(C)f^D+imag(C)f^E+F\{1-(f/G)^2-2j*(f/H)\}^{-1}$$

	B	C	D	E	F	G	H
$\varepsilon_2(f)$	2.9+0.39j	1.01–0.69j	0.51+0.32j	0.32–2E-002j	2.24+0.24j	7.49–2.7j	–8.4–2.6j
500-nm Fe 13.4% + 80-µm $Ni_{.32}Zn_{.48}Cu_{.12}Fe_2O_{4.6}$ at 13.6%/volumes in epoxy, 2.43 g/cc, 0.05–10 GHz							
$\chi_m(f)$	2.85+1.05j	0.56+1.78j	0.4+5E-002j	0.99+2.11j			
$\|B\|$, $\|C\|$ and $\|D\|$	3.04	1.87	0.403				
$\chi_{DCP}, f_{rc}, f_{cd}, f_{pd}$	143	1.87	0.31	0.403	0.011		
$\varepsilon_1(f)$	1.4+1.37j	2.76–1.06j	2E-002j	2.45+2.9j	266.8–266.9j	–24.8–26.1j	
$\varepsilon_2(f)$	3.53+0.2j	0.92–0.75j	0.29+0.14j	0.44+4E-002j	2.92+0.2j	11.6–12j	–34.8+6.3j
500-nm Fe 17.5% + 80-µm $Ni_{.32}Zn_{.48}Cu_{.12}Fe_2O_{4.6}$ at 17.5%/volumes in epoxy, 2.82 g/cc, 0.05–10 GHz							
$\chi_m(f)$	4.82+0.64j	0.83–0.47j	–0.29+0.22j	–1.3–0.2j			
$\|B\|$, $\|C\|$ and $\|D\|$	4.86	0.954	0.3640				
$\chi_{DCP}, f_{rc}, f_{cd}, f_{pd}$		0.954		0.364			
$\varepsilon_1(f)$	2.08–0.82j	3.16–0.69j	0.28+9E-002j	5.78+2.28j	4.2+0.3j	1.26–2.32j	
$\varepsilon_2(f)$	4.3+0.26j	1.13–0.86j	–0.18–0.1j	0.57+0.11j	3.47+0.25j	2–16.1j	–0.32+22.9j
Iron–Ferrite Composites 0.5							
500-nm Fe 19.9% + 80-µm $Ni_{.32}Zn_{.48}Cu_{.12}Fe_2O_{4.6}$ at 20.0% by volumes in epoxy, 3.1 g/cc, 0.001–10 GHz							

Sample ID and Information

$$\mu(f)-1=\chi_m(f)=B(1-j*f/D)\{1-(f/C)^2-j*(f/D)\}^{-1}$$

$\chi_{DCP}, f_{rc}, f_{rp}, f_{cd}, f_{pd}$ are, respectively, the EMT-calculated DC magnetic particle susceptibility; calculated composite resonant frequency, |*C*|; EMT-calculated magnetic particle resonant frequency; calculated composite relaxation frequency, |*D*|; EMT-calculated magnetic particle relaxation frequency. Note, please apply the absolute parameter values (i.e., |*B*|, |*C*|, |*D*|) when calculating permeability.

$$\varepsilon_1(f)=B+Cf^D+E\{1-(f/F)^2-2j*(f/G)\}^{-1}$$
$$\varepsilon_2(f)=B+real(C)f^D+imag(C)f^E+F\{1-(f/G)^2-2j*(f/H)\}^{-1}$$

	B	C	D	E	F	G	H
$\chi_m(f)$	8.14+0.13j	0.54–0.1j	–0.21+2E-002j	–1.29+0.12j			
|*B*|, |*C*| and |*D*|	8.14	0.55	0.21				
$\chi_{DCP}, f_{rc}, f_{cd}, f_{pd}$	100	0.55	0.12	0.21	1.3		
$\varepsilon_1(f)$	2.41–1.8j	3.27–1.2j	8E-002+0.61j	6.13+6.3j	1.39+1.64j	0.74–0.86j	
$\varepsilon_2(f)$	3.67–4E-002j	0.75–0.53j	–0.3–0.4j	–0.21+0.11j	3.05–6E-002j	23.64+12.3j	16.3+5.2j
500-nm Fe 23.3% + 80-µm $Ni_{.32}Zn_{.48}Cu_{.12}Fe_2O_{4.6}$ at 23.3%/volumes in epoxy, 3.4 g/cc, 0.05–10 GHz							
$\chi_m(f)$	8.41+0.16j	0.52-7E–002j	–0.19-2E–004j	–1.34+0.21j			
|*B*|, |*C*| and |*D*|	8.41	0.525	0.19				
$\chi_{DCP}, f_{rc}, f_{cd}, f_{pd}$	60.1	0.525	0.167	0.19	1.36		
$\varepsilon_1(f)$	5.41+1.1j	4-0.79j	–2E-003–2.6E-002j	–0.61+0.7j	0.18+0.57j	0.12+7E-002j	
$\varepsilon_2(f)$	4.48+0.16j	1.69–0.46j	–0.2–7E-002j	–0.62+0.11j	3.65+8E-002j	29.6–23.5j	–115+53.4j

Figure 12.6 *Data show measurements and fit to a three phase magnetic composite. Model and measurement have near overlaps.*

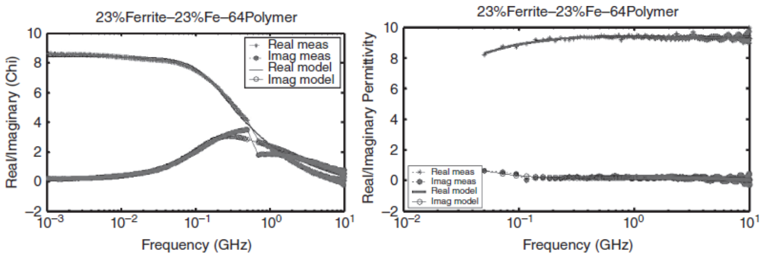
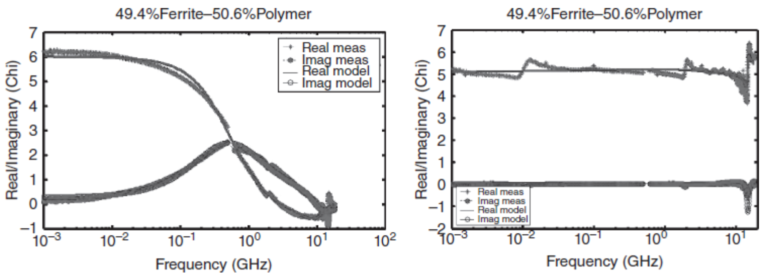


Figure 12.7 *Data show measurements and fit to a two phase magnetic composite. Model and measurement have near overlaps.*



Disclaimer: Data are results obtained by the author on samples prepared in the laboratory. Variations may be observed by other researchers when prepared in their laboratories. Data are not "guaranteed" by a manufacturer or the author. Variations in properties may be observed and users of the products should verify properties. All material properties are guides for engineering design. Materials prepared in the laboratory should be considered as experimental and also guides for engineering design.